Question Number etc. in left margin | Mark allocation in right margin

1a)

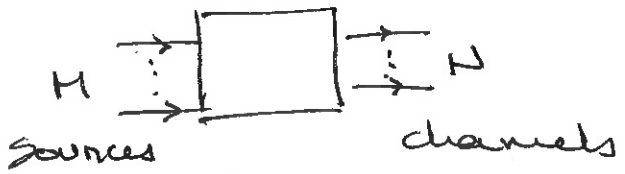
i) Each idle source is a Poisson source:

$P(\text{new demand in } (t, t+\Delta t) \mid \text{source idle}) = \lambda \Delta t$

channel holding time are exponentially distributed with mean $1/\mu$

full availability

$M \sim N$



note: j channels busy $\leftrightarrow$ j sources busy

ii) so, the total arrival rate to the link will fall as N_t increases. 2

$\lambda_i = (M-i)\lambda = (1 - i/M)\lambda_0 \quad \lambda_0 = M\lambda \quad i < i_{max}$

$\mu_i = i\mu \quad i > 0$ 2

iii) when $M \gg N$ the arrival rate in state i is $\lambda_i = (1 - i/M)\lambda_0 \sim \lambda_0$ for all i

Erlang: the total arrival stream (of demands) is a Poisson process with parameter λ which is not affected by the state of the system i 2

Question Number etc. in left margin | Mark allocation in right margin

1b) i) **Mean $E(N_t)$**

$$\sum_{j=0}^{n} j \frac{\rho^j}{j!} P_0 = \sum_{j=0}^{n} \frac{j\rho^j}{j(j-1)!} P_0 = \rho \sum_{j=0}^{n} \frac{\rho^{j-1}}{(j-1)!} P_0 \quad 3$$

$$E(N_t) = \rho \sum_{i=0}^{n-1} \frac{\rho^i}{i!} P_0 = \rho \left(1 - \frac{\rho^n}{n!} P_0\right) = \rho(1 - B_c) \quad 3$$

ii) **Variance $Var(N_t)$**

$$E(N_t^2) = \sum_{j=0}^{n} j^2 \frac{\rho^j}{j!} P_0 = \rho \sum_{j=0}^{n} \frac{j\rho^{j-1}}{(j-1)!} P_0 \quad 2$$

$$= \rho \sum_{i=0}^{n-1} \frac{(i+1)\rho^i}{i!} P_0 = \rho \left[\underbrace{\sum_{i=0}^{n-1} \frac{i\rho^i}{i!} P_0}_{} + \underbrace{\sum_{i=0}^{n-1} \frac{\rho^i}{i!} P_0}_{1 - B_c} \right] \quad 2 \quad 2$$

$$+ \frac{n\rho^n}{n!} P_0 - \frac{n\rho^n}{n!} P_c$$

$$= \rho(1 - B_c) - n B_c \quad 2$$

$$E(N_t^2) = \rho \left[\rho(1 - B_c) - n B_c + (1 - B_c)\right]$$

$$Var(N_t) = E(N_t^2) - (E(N_t))^2$$

$$= \rho^2 - \rho^2 B_c - n\rho B_c + \rho - \rho B_c$$

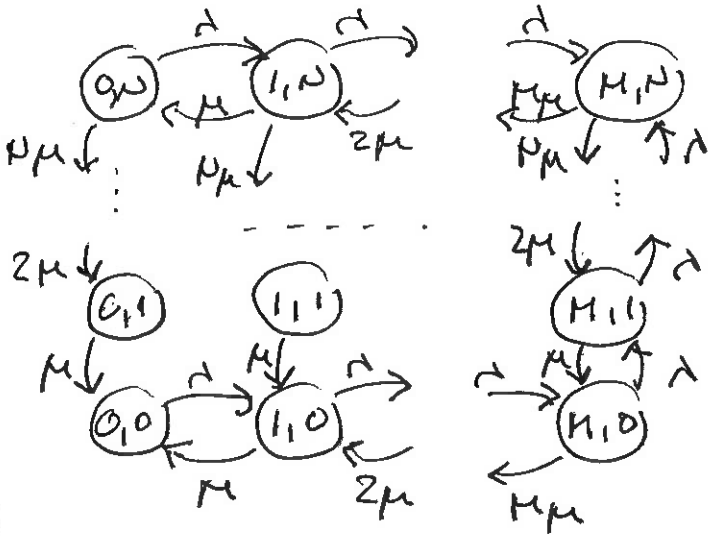
$$- \rho^2 + 2 B_c \rho^2 - \rho^2 B_c^2$$

$$= \rho - \rho B_c - \rho n B_c + \rho^2 B_c - \rho^2 B_c^2$$

$$\rho(1 - B_c) - \rho B_c [n - (n - B_c)\rho]$$

$$Var(N_t) = E(N_t) - \rho B_c [n - (1 - B_c)\rho] \quad 2$$

2a i)



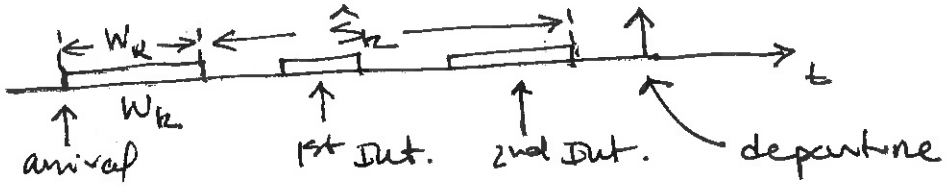
3

3

ii) 2-D Birth/death model for overflow traffic
link 1: M channels link 2: N channels
offered traffic is pure chance traffic

2

2b)



2

$\hat{S}_k$ = effective service time

V_k = work brought into system during $\hat{S}_k$ by higher priority arrivals

2

$$E[V_k] = \sum_{i=1}^{k-1} \lambda_i E[\hat{S}_k] E[S_i] = \left(\sum_{i=1}^{k-1} \rho_i\right) E[\hat{S}_k]$$ 2

$$= \sigma_{k-1} E[\hat{S}_k]$$ (using $\rho_i = \lambda_i E[S_i]$) 2

$$E[\hat{S}_k] = E[S_k] + E[V_k] = E[S_k] + \sigma_{k-1} E[\hat{S}_k]$$ 2

$$E[\hat{S}_k] = \frac{E[S_k]}{1 - \sigma_{k-1}}$$ 2

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3a offered traffic $\frac{180 \times 180}{3600} = 9$ Erlangs

i) P[overflow channel is blocked] = occupancy of channel

$= \rho_0 [E_{15}(\rho_0) - E_{16}(\rho_0)] = 9[0.020 - 0.011]$

$= 0.081$ 4

ii) IPP model (mean and variance of traffic)

Mean $M = \rho_0 E_{15}(\rho_0) = 0.18 = m$

Variance $V = m\left(1 - m + \frac{\rho_0}{M + m + 1 - \rho_0}\right) = 0.373$ $M = 15$ 4

For the IPP model use $\mu(k)$

$$E(N_t) = \frac{\alpha}{\alpha + \mu} \rho_0$$

$$E(N_t(N_t - 1)) = \left(\frac{\alpha}{\alpha+\mu}\right)\left(\frac{\alpha+1}{\alpha+\mu+1}\right)\rho_0^2$$

$$= \text{Var}[N_t] + [E(N_t)]^2 - E(N_t)$$

$$= 0.2254$$ 4

$\alpha = 0.145$
$\beta = 7.08$ } 2

3b In this system for delayed arrivals we have:

$$P(w>2 \mid Q_t=i) = P(<(i+1) \text{ departures in } (0,2)) = \sum_{j=0}^{i} \frac{(K\mu 2)^j}{j!} e^{-K\mu 2}$$ 2

Therefore for delay arrivals

$$P(w>2) = \sum_{i=0}^{\infty} P(Q_t=i)\, P(w>2 \mid Q_t=i)$$

in equilibrium for delay arrivals

$$P(Q_t=i) = (1-\rho)\rho^i$$ hence

$$P(w>2) = \sum_{i=0}^{\infty} (1-\rho)\rho^i \sum_{j=0}^{i} \frac{(K\mu 2)^j}{j!} e^{-K\mu 2}$$ 2

$$(1-\rho) + (1-\rho)\rho\left(1 + \frac{K\mu 2}{1!}\right) + (1-\rho)\rho^2\left(1 + \frac{K\mu 2}{1!} + \frac{(K\mu 2)^2}{2!}\right) + \ldots$$

$$= (1)\left[(1-\rho) + (1-\rho)\rho + (1-\rho)\rho^2 \ldots\right] +$$

$$\rho\frac{K\mu 2}{1!}\left[(1-\rho) + (1-\rho)\rho + (1-\rho)\rho^2 \ldots\right] +$$

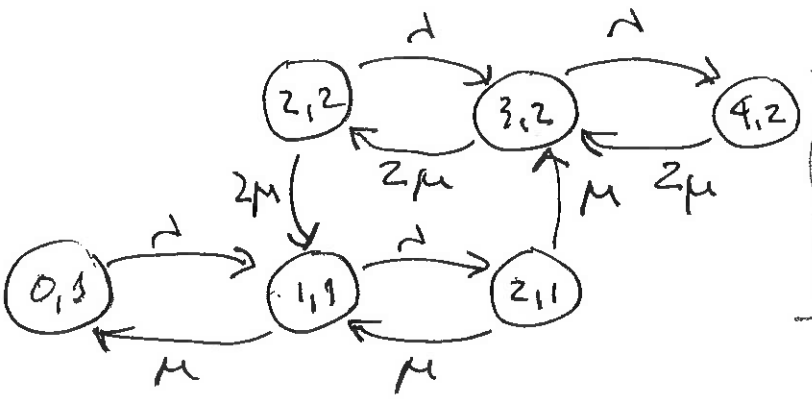
$$\rho^2\frac{(K\mu 2)^2}{2!}\underbrace{\left[(1-\rho) + (1-\rho)\rho + (1-\rho)\rho^2 \ldots\right]}_{=1} + \ldots$$

$$= \sum_{i=0}^{\infty} \frac{(K\mu 2\rho)^i}{i!} = e^{K\mu 2\rho}$$

$$P(w>2) = e^{K\mu 2\rho} \cdot e^{-K\mu 2} = e^{-K\mu(1-\rho)2}$$ 2

Question Number etc. in left margin | Mark allocation in right margin

4a

i) 

(i,j) state space 2

rates ⟷ 2 — 2

2 states — 2

rates ↕ 2 — 2

ii) $R = [0, 1, 2, 4, 6, 8]$ — 2

4b

$$\varepsilon = \int_{J_0}^{\infty} \frac{e^{-(x-m)^2/2\sigma^2}}{\sqrt{2\pi\sigma^2}}\, dx$$ — 4

iii) $$\varepsilon = \int_{c}^{\infty} \frac{e^{-\frac{(x-m)^2}{2\sigma^2}}}{\sqrt{2\pi\sigma^2}}\, dx$$

$$y = \frac{x-m}{\sqrt{2}\sigma}$$

$$dy = \frac{dx}{\sqrt{2}\sigma}$$ — 2

$$\varepsilon = \frac{1}{\sqrt{2\pi\sigma^2}} \int_{c}^{\infty} e^{-y^2} \sqrt{2}\sigma\, dy = \frac{\sqrt{2}\sigma}{\sqrt{2\pi}\sigma} \int_{c}^{\infty} e^{-y^2} dy$$ — 2

$$\varepsilon = \frac{\sqrt{2}\sigma}{\sqrt{\pi}} \frac{e^{-\frac{(c-m)^2}{2\sigma^2}}}{2(c-m)} = \frac{\sigma e^{-\frac{(c-m)^2}{2\sigma^2}}}{\sqrt{2\pi}(c-m)}$$

$$\varepsilon = \frac{\sigma e^{-\frac{(c-m)^2}{2\sigma^2}}}{\sqrt{2\pi}(c-m)}$$ — 2